

Limitations of the Sequential Parallel Comparisons Design

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Outline

I. The Design

II. Estimands & Estimators

II. Assumptions

III. Misclassification

IV. Discussion

The Aim of the SPCD

The Goal

The SPCD aims to reduce the overall placebo response rate as well as the required sample size by identifying placebo non-responders and re-randomizing these participants in a second stage.

Diagram

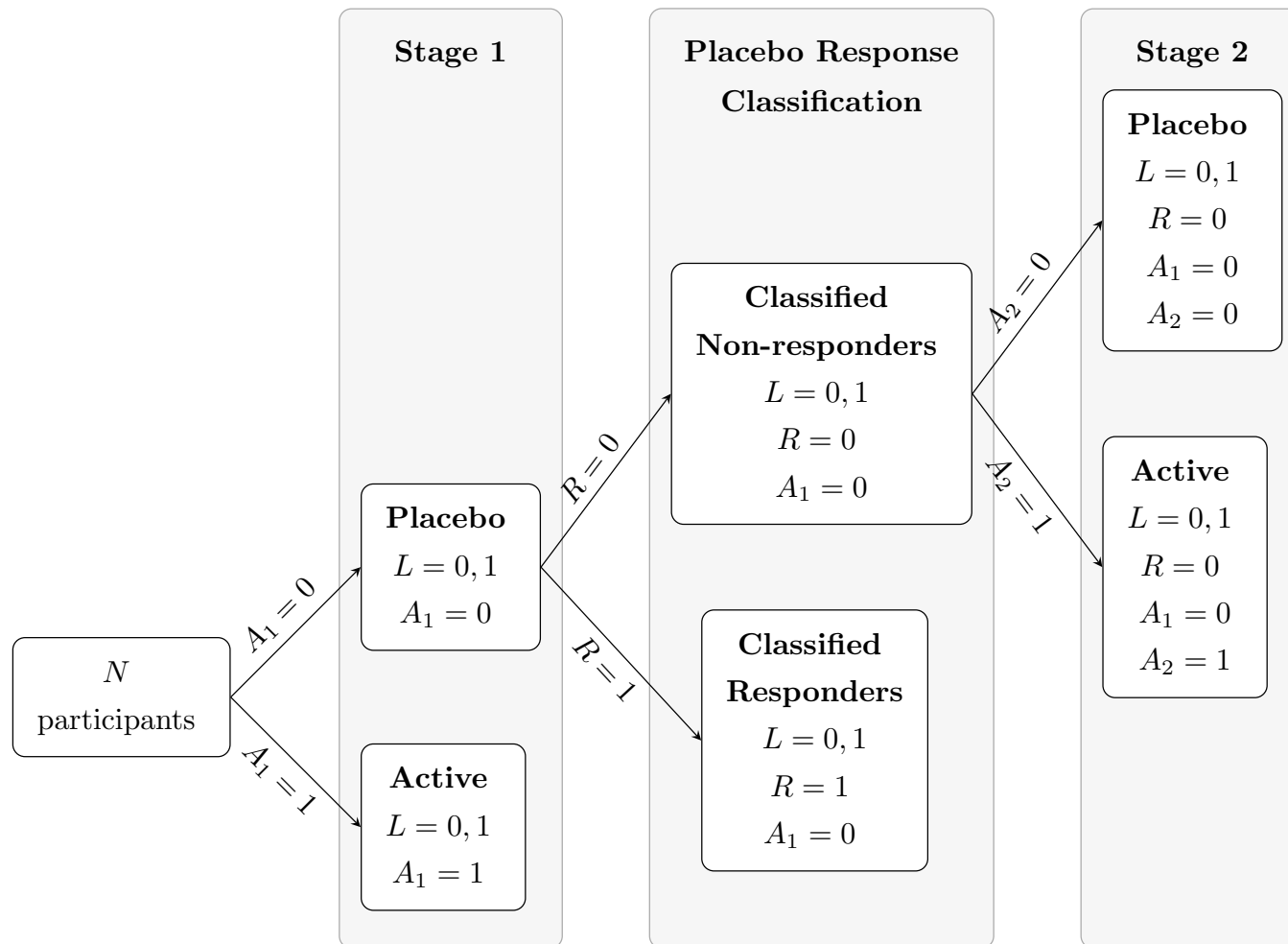


Figure 1: An overview of the Sequential Parallel Comparisons Design (SPCD) [1].

Simulated Example

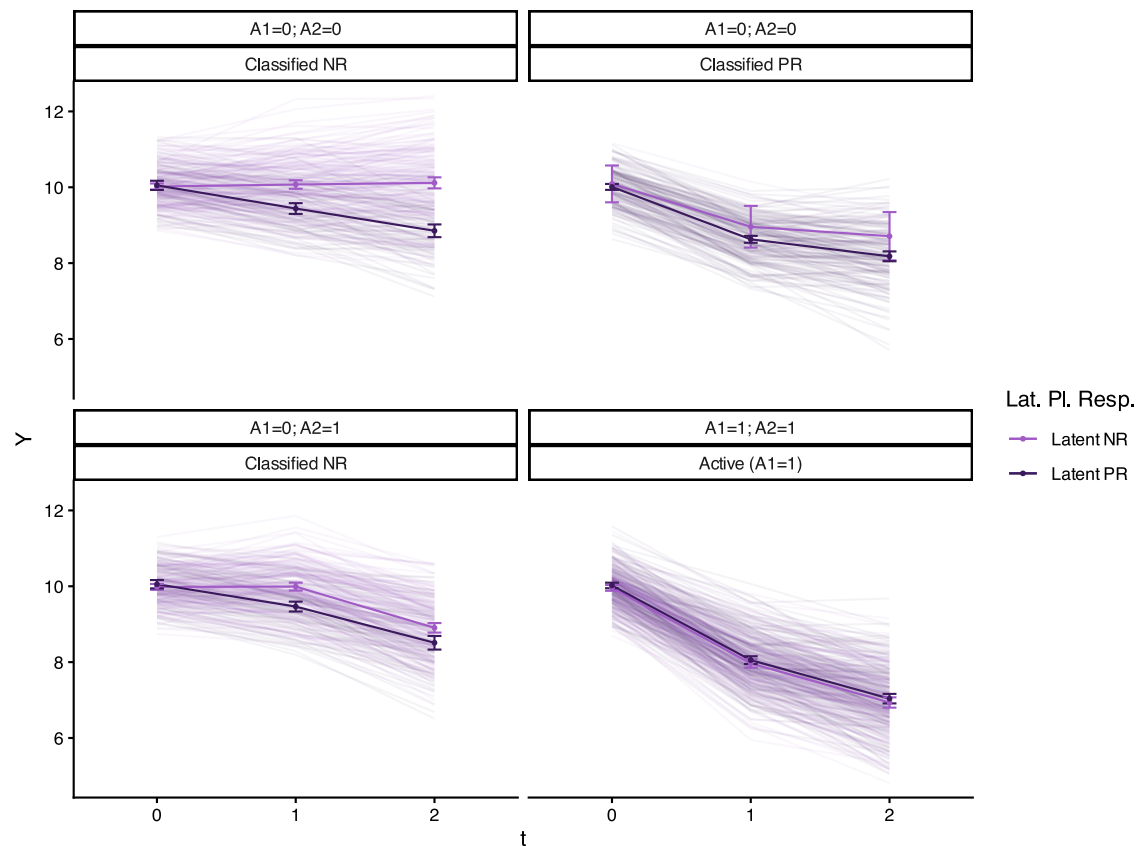


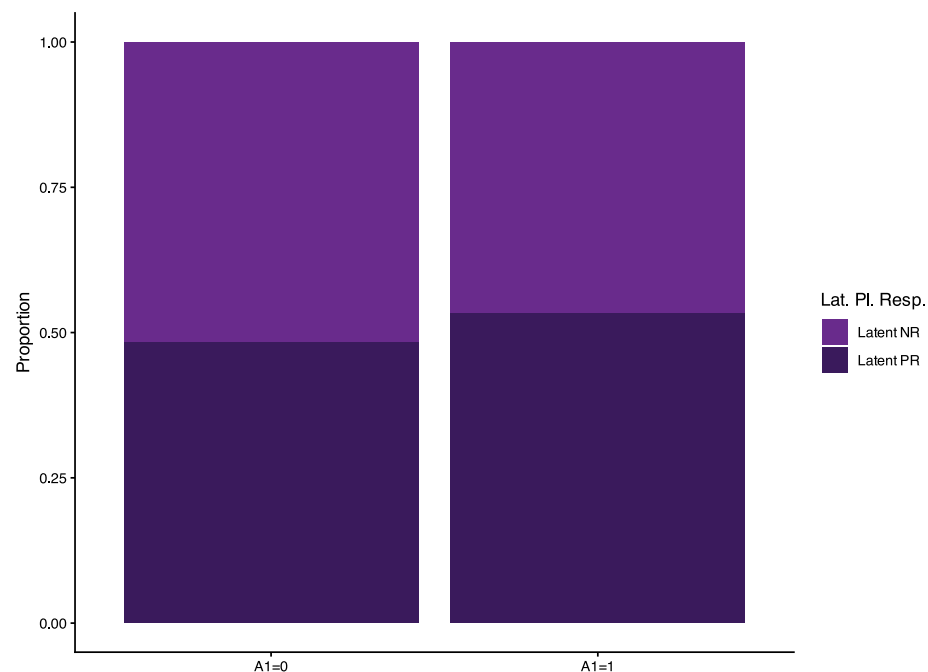
Figure 2: Spaghetti plot of simulated SPCD data color-coded by latent placebo response status (L).

Stage 1

- Analogous to a Parallel Arm Randomized Control Trial
- Placebo treated arm is a latent mixture of placebo responders ($L = 1$) and non-responders ($L = 0$)
- Natural contrast is between Active ($A_1 = 1$) and Placebo ($A_1 = 0$) Treatments

$$Y_1(A_1 = 1) - Y_1(A_1 = 0)$$

$$\hat{\theta}_1 = \hat{E}(Y_1(1) \mid Y_0) - \hat{E}(Y_1(0) \mid Y_0)$$



Mixture of latent placebo response status (L) in Stage 1 treatment arms.

Stage 2

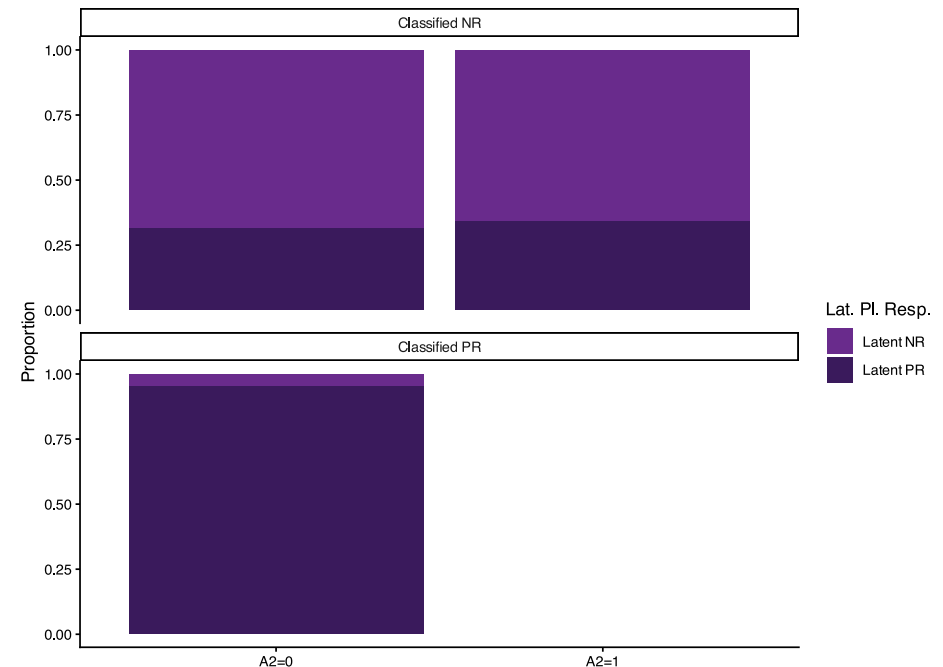
- After Stage 1, placebo participants are classified as responders

($R = 1$) or non-responders ($R = 0$)

- Classified** non-responders will inevitably contain some **true** placebo responders due to *misclassification*
- Natural contrast is between Active and Placebo for classified placebo non-responders

$$Y_2(A_2 = 1) - Y_2(A_2 = 0) \mid R = 0, A_1 = 0$$

$$\begin{aligned} \hat{\theta}_2 = & \hat{E}(Y_2(1) \mid R = 0, A_1 = 0, Y_1, Y_0) \\ & - \hat{E}(Y_2(0) \mid R = 0, A_1 = 0, Y_1, Y_0) \end{aligned}$$



Mixture of latent placebo response status (L) in Stage 2 treatment arms sorted by classified response status.

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Estimands

Quantity	Population Subset	Formula
Δ_{placebo}	Stage 1 Placebo; $A_1 = 0$	$E(Y_1(0) \mid L = 1) - E(Y_1(0) \mid L = 0)$
Δ_{NR}	Placebo NR; $L = 0$	$\mathbb{E}(Y_1(1) \mid L = 0) - E(Y_1(0) \mid L = 0)$
Δ_{PR}	Placebo Resp.; $L = 1$	$\Delta_{NR} - \Delta_{\text{placebo}}$
Δ_{all}	All Participants	$\Delta_{NR} - P(L = 1)\Delta_{\text{placebo}}$

Table 1: Average effects as functions of counterfactual outcomes and latent placebo responder status.

Estimators

Estimator	Targeted Estimand	Formula
$\hat{\theta}_1$	Δ_{all}	$\hat{E}(Y_1(1) Y_0) - \hat{E}(Y_1(0) Y_0)$
$\hat{\theta}_2$	Δ_{NR}	$\hat{E}(Y_2(1) R = 0, Y_1, Y_0) - \hat{E}(Y_2(0) R = 0, Y_1, Y_0)$
$\hat{\theta}_w$	?	$w\hat{\theta}_1 + (1 - w)\hat{\theta}_2$

Table 2: Conventional estimators for the SPCD.

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Conventional Identification Assumptions

Consistency/SUTVA

$$Y_1 = Y_1(A_1) \mid L \quad \text{and} \quad Y_{i,1}(A_{i,1} = a) \perp A_{j,1} \mid L = l \quad \text{for all } i \neq j.$$

Ignorability

$$A_1 \perp Y_1(a_1) \mid L = l \quad \text{for all } a_1, l.$$

Positivity

$$0 < P(A_1 = a) < 1 \quad \text{for all } a.$$

Implausible, but Required Assumptions

Latent-type Stability

$$L \perp A_1$$

Effect Homogeneity

$$E(Y_t(A_t = 1) \mid L = 1) = E(Y_t(A_t = 1) \mid L = 0)$$

Mixture Identifiability

Given that Y_1 follows a mixture, let $p_L = P(L = 1)$ be the unknown placebo responder prevalence, $f_{Y_1}(\cdot; \xi)$ be component densities from family $\mathcal{F} = \{f(\cdot; \xi)\}$ indexed by ξ_l for latent response status $L = l$. Then the density of the stage 1 outcome in the placebo arm is,

$$f_{Y_1|A_1=0}(y) = p_L f_{Y_1}(y; \xi_1) + (1 - p_L) f_{Y_1}(y; \xi_0),$$

such that the mixture is identified under that parametric model.

Estimator Expectations

Estimator	Targeted Estimand	Actual Expectation
$\hat{\theta}_1$	Δ_{all}	Δ_{all}
$\hat{\theta}_2$	Δ_{NR}	$\Delta_{NR} - p_{L \mid R=0} \Delta_{\text{placebo}}$
$\hat{\theta}_w$	?	$\Delta_{NR} - \Delta_{\text{placebo}}[wp_L + (1 - w)p_{L \mid R=0}]$

Table 3: Expectations of the conventional estimators. Note, $p_L = P(L = 1)$ and $p_{L \mid R=0} = P(L = 1 \mid R = 0)$.

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Misclassification is Inevitable...

- A limitation of previous work on SPCD is the absence of substantial discussion of placebo response classification, with some recent exceptions[2], [3], [4], [5].

$$C_R(y_0, y_1) = I(y_1 - y_0 > c) \text{ or } C_R(y_0, y_1) = I(y_1 > c), \quad (1)$$

- If placebo responders are misclassified as non-responders, then the estimate for Δ_{NR} is contaminated
- Responder misclassification is inevitable unless $C_R(y_0, y_1) = 1$ for all participants (no non-responders)

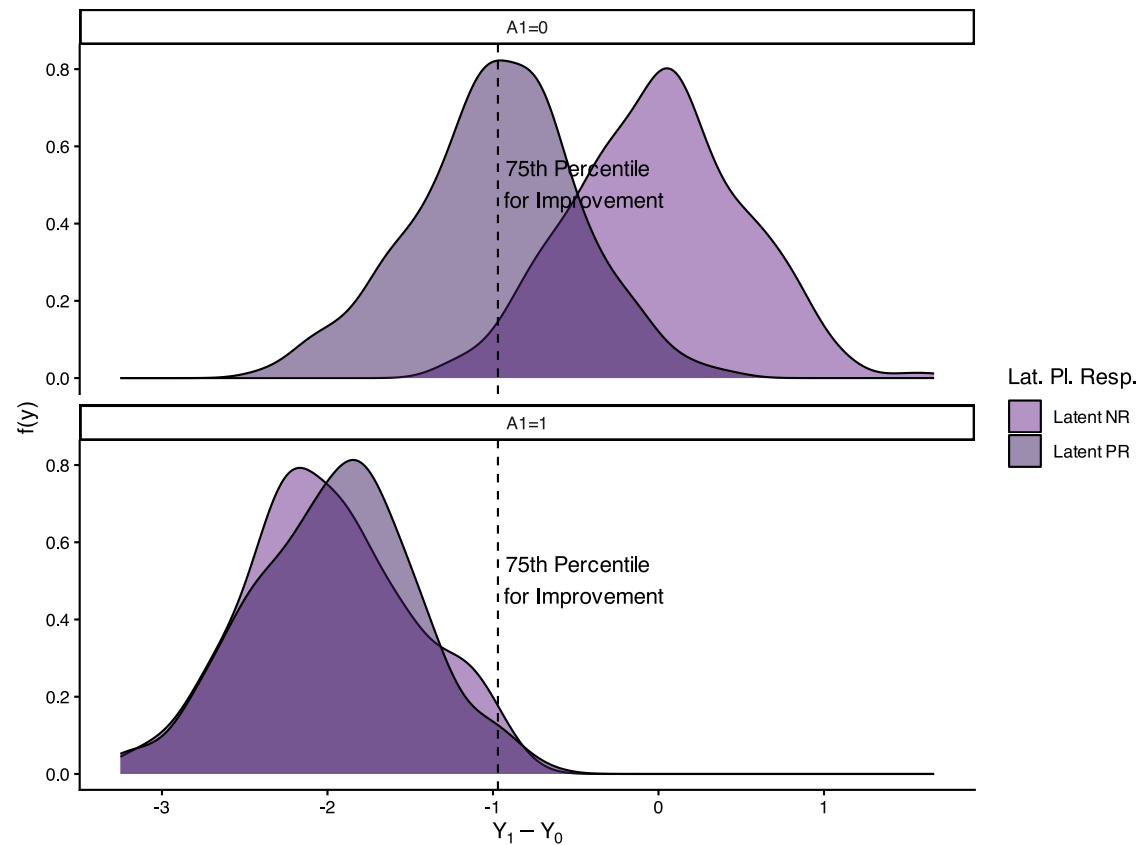


Figure 3: Distributions of individual responses after stage 1 by Stage 1 Treatment & Latent placebo response status.

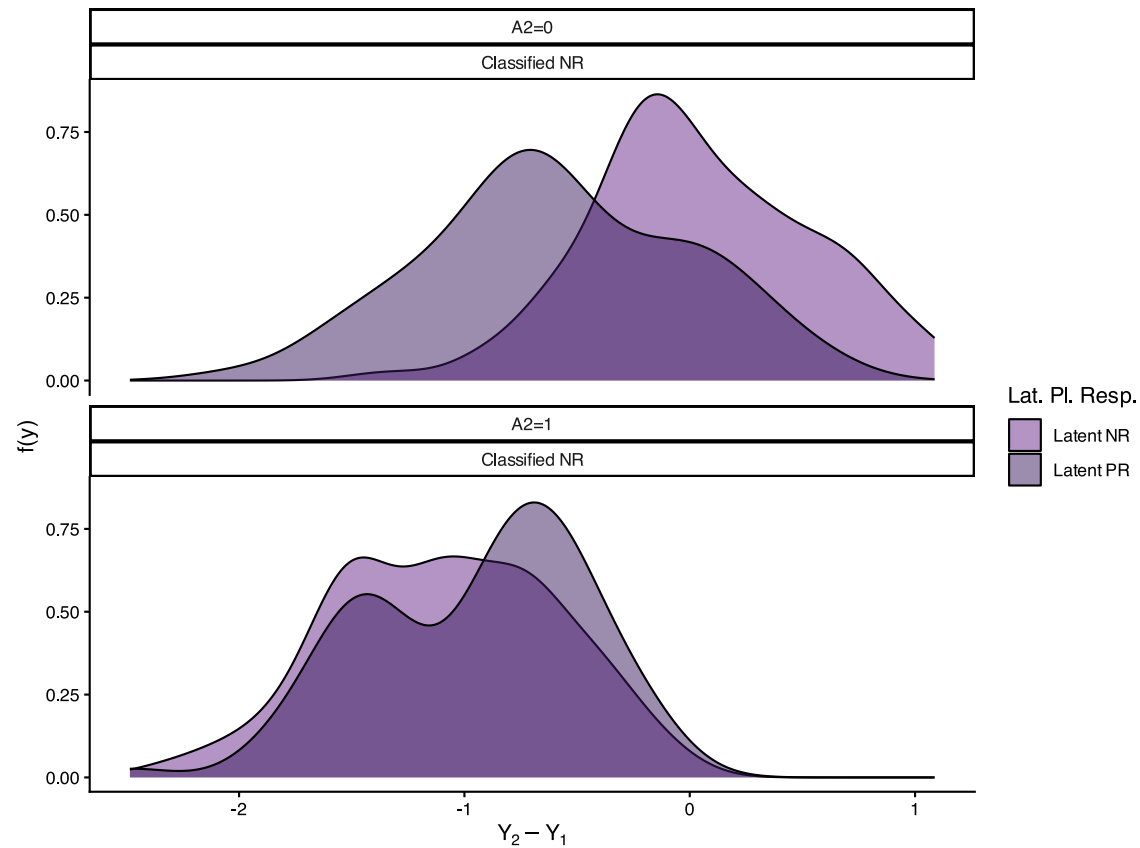


Figure 4: Distributions of individual responses after stage 2.

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Take-aways

- SPCD has several implausible assumptions baked into the design
- Misclassification makes the estimation unreliable even if you are willing to make the assumptions
- The conventional Stage 2 and weighted estimands don't target anything meaningful and are difficult to interpret
- Alternatives ...
 - Latent regression with a continuous placebo
 - Optimal partitioning via finite mixture models
 - Placebo lead-in design
 - Conventional crossover design

Warning

Don't use the SPCD!

Acknowledgements

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Computation

The computational work performed on this project was done with help from the NYU Big Purple High-Performance Computing cluster.

Questions?

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